

# Sum\_functions\_debug\_041824\_documentation

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Documentation for Sum\_functions\_debug\_041824.r. This file contains functions used to sum by subgroups, used in the code that calculates joint effects.

Many are the exact same as defined in “Sum.by.strata.redone.r”, redefined here because they’ve been removed early in the “Part3\_debug\_041824.R” code.

```

# only works on specific allmort file created by my code
# use covar="overall" for all

# Functions -----

# function of calculating RE PAFs sum stats
get.RE.PAFs.draws<-function(allmort, totalmort) ##for now, only works for age/sex/race
{
  merged<-merge(allmort, totalmort)

  allmort.starting.point<-which(names(merged)== "V1")
  allmort.ending.point<-which(names(merged)== "riskfactor")-1
  totalmort.starting.point<-which(names(merged)== "X1")
  totalmort.ending.point<-dim(merged)[2]

  RE.PAFs.info<-merged[,which(names(merged) %in% c("age", "female", "race", "outcome", "mean..food.", "se..f

  RE.PAFs<-cbind(merged[,which(names(merged) %in% c("age", "female", "race", "outcome", "mean..food.", "se..f
    merged[,allmort.starting.point:allmort.ending.point])/merged[,totalmort.starting.point:total
  RE.PAFs<-RE.PAFs[order(RE.PAFs$riskfactor, RE.PAFs$outcome, RE.PAFs$female, RE.PAFs$age, RE.PAFs$race),]

  return(RE.PAFs)
}

get.RE.PAFs<-function(allmort, totalmort) ##for now, only works for age/sex/race
{
  merged<-merge(allmort, totalmort)

  allmort.starting.point<-which(names(merged)== "V1")
  allmort.ending.point<-which(names(merged)== "riskfactor")-1
  totalmort.starting.point<-which(names(merged)== "X1")
  totalmort.ending.point<-dim(merged)[2]

  RE.PAFs.info<-merged[,which(names(merged) %in% c("age", "female", "race", "outcome", "mean..food.", "se..f

  RE.PAFs<-cbind(merged[,which(names(merged) %in% c("age", "female", "race", "outcome", "mean..food.", "se..f
    merged[,allmort.starting.point:allmort.ending.point])/merged[,totalmort.starting.point:total
  RE.PAFs<-RE.PAFs[order(RE.PAFs$riskfactor, RE.PAFs$outcome, RE.PAFs$female, RE.PAFs$age, RE.PAFs$race),]

  RE.PAFs.sum.stats<-sum.stats.maker.RE.PAFs(RE.PAFs, n.covar)
  return(RE.PAFs.sum.stats)
}

# function to get summary stats
sum.stats.maker<-function(allmort)
{
  allmort.dt<-as.data.table(allmort)
  vars<-paste("V", 1:n.sims, sep="")

  mort.summary <- allmort.dt[, "!="(means = rowMeans(.SD, na.rm = TRUE),
    sd.devs = apply(.SD, 1, sd),
    medians = apply(.SD, 1, median),
    LB = apply(.SD, 1, function(x) quantile(x, .025, na.rm = T)),
    UB = apply(.SD, 1, function(x) quantile(x, .975, na.rm = T))
  ),
  .SDcols = vars

```

```

]

mort.summary <- mort.summary[, (vars) := NULL]

return(mort.summary)
}

sum.stats.maker.RE.PAFs<-function(RE.PAFs)
{
  RE.PAFs.dt<-as.data.table(RE.PAFs)
  vars<-paste("V", 1:n.sims, sep="")

  RE.PAFs.summary <- RE.PAFs.dt[, "!="(RE.PAF.means = rowMeans(.SD, na.rm = TRUE),
                                     RE.PAF.sd.devs = apply(.SD, 1, sd),
                                     RE.PAF.medians = apply(.SD, 1, median),
                                     RE.PAF.LB = apply(.SD, 1, function(x) quantile(x, .025, na.rm = T)),
                                     RE.PAF.UB = apply(.SD, 1, function(x) quantile(x, .975, na.rm = T)),
                                     RE.PAF.medians.as.percent = apply(.SD, 1, median)*100,
                                     RE.PAF.LB.as.percent = apply(.SD, 1, function(x) quantile(x, .025, na.rm = T)),
                                     RE.PAF.UB.as.percent = apply(.SD, 1, function(x) quantile(x, .975, na.rm = T))),
    .SDcols = vars
  ]

  RE.PAFs.summary <- RE.PAFs.summary[, (vars) := NULL]

  return(RE.PAFs.summary)
}

# debug
# allmort=allmort
# totalmort=totalmort
# pop=pop.draws
# i=36
# covar=strata.combos[[i]]

Sum.by.strata<-function(allmort, totalmort, pop, covar)
{
  allmort.dt<-as.data.table(allmort)
  totalmort.dt<-as.data.table(totalmort)
  pop.dt<-as.data.table(pop)

  strata<-covar

  cols.to.sum.allmort<-paste("V", 1:n.sims, sep="")
  cols.to.sum.totalmort<-paste("X", 1:n.sims, sep="")
  cols.to.sum.pop<-paste("Y", 1:n.sims, sep="")

  strata.sims.allmort<-allmort.dt[,lapply(.SD, sum), by=c(strata, "riskfactor"), .SDcols=cols.to.sum.allmort]
  strata.sims.totalmort<-totalmort.dt[,lapply(.SD, sum), by=c(strata[strata %in% c("age", "female", "race"),
                                     .SDcols=cols.to.sum.totalmort]
  strata.sims.pop<-pop.dt[,lapply(.SD, sum), by=c(strata[strata %in% c("age", "female", "race")]), .SDcols=cols.to.sum.pop]

```

```

#merged<-reduce(list(strata.sims.allmort, strata.sims.totalmort, strata.sims.pop), merge)
if(length(strata[strata %in% c("age", "female", "race")])>0 & dim(strata.sims.totalmort)[1]>1)
  merged<-merge(merge(strata.sims.allmort, strata.sims.totalmort), strata.sims.pop, by=strata[strata %in%
if(length(strata[strata %in% c("age", "female", "race")])>0 & dim(strata.sims.totalmort)[1]==1)
{
  merge.a<-cbind(strata.sims.allmort, strata.sims.totalmort)
  merged<-merge(merge.a, strata.sims.pop, by=strata[strata %in% c("age", "female", "race")])
}
if(length(strata[strata %in% c("age", "female", "race")])==0 & dim(strata.sims.totalmort)[1]>1)
{
  merged<-cbind(merge(strata.sims.allmort, strata.sims.totalmort), strata.sims.pop)
}
if(length(strata[strata %in% c("age", "female", "race")])==0 & dim(strata.sims.totalmort)[1]==1)
  merged<-cbind(strata.sims.allmort, strata.sims.totalmort, strata.sims.pop)

allmort.starting.point<-which(names(merged)=="V1")
allmort.ending.point<-which(names(merged)==paste("V", nsim1, sep=""))
totalmort.starting.point<-which(names(merged)=="X1")
totalmort.ending.point<-which(names(merged)==paste("X", nsim1, sep=""))
pop.starting.point<-which(names(merged)=="Y1")
pop.ending.point<-which(names(merged)==paste("Y", nsim1, sep=""))

#RE.PAFs.info<-merged[,c(unlist(strata.combos[j]), "outcome"), with=FALSE]
RE.PAFs<-cbind(merged[,c(strata, "riskfactor"), with=FALSE],
  merged[,allmort.starting.point:allmort.ending.point]/merged[,totalmort.starting.point:total

setorderv(RE.PAFs, cols=c("riskfactor", strata))

sum.stats.attr.mort.PAF<-sum.stats.maker(strata.sims.allmort)
RE.PAFs.sum.stats<-sum.stats.maker.RE.PAFs(RE.PAFs)

standardized.mort<-cbind(merged[,c(strata, "riskfactor"), with=FALSE],
  merged[,allmort.starting.point:allmort.ending.point]/merged[,pop.starting.point:pop

setorderv(standardized.mort, cols=c("riskfactor", strata))
standardized.mort.sum.stats<-sum.stats.maker(standardized.mort)
names(standardized.mort.sum.stats)[(length(names(standardized.mort.sum.stats))-4):length(names(standardize
  paste("mort.per.100k.", names(standardized.mort.sum.stats)[(length(names(standardized.mort.sum.stats))-4)

#merged.results<-merge(sum.stats.attr.mort.PAF, RE.PAFs.sum.stats)
merged.results<-reduce(list(sum.stats.attr.mort.PAF, RE.PAFs.sum.stats, standardized.mort.sum.stats), merg

strata.summary.stats<-merged.results

return(list(strata.summary.stats, strata.sims.allmort, strata.sims.totalmort, RE.PAFs, standardized.mort))
}

Sum.by.overall<-function(allmort, totalmort, pop)
{
  allmort.dt<-as.data.table(allmort)
  totalmort.dt<-as.data.table(totalmort)
  pop.dt<-as.data.table(pop)

  cols.to.sum.allmort<-paste("V", 1:n.sims, sep="")
  cols.to.sum.totalmort<-paste("X", 1:n.sims, sep="")

```

```

cols.to.sum.pop<-paste("Y", 1:n.sims, sep="")

strata.sims.allmort<-allmort.dt[,lapply(.SD, sum), by=c("outcome", "disease_type", "riskfactor"), .SDcols=
strata.sims.totalmort<-totalmort.dt[,lapply(.SD, sum), by=c("outcome", "disease_type"), .SDcols=cols.to.sum
strata.sims.pop<-pop.dt[,lapply(.SD, sum), .SDcols=cols.to.sum.pop]

#merged<-merge(strata.sims.allmort, strata.sims.totalmort)
#merged<-reduce(list(strata.sims.allmort, strata.sims.totalmort, strata.sims.pop), merge)
#merged<-merge(merge(strata.sims.allmort, strata.sims.totalmort), strata.sims.pop, by="")
merged<-merge(strata.sims.allmort, strata.sims.totalmort)
#replicate to get n rows of same numbers
strata.sims.pop<-map_dfr(seq_len(dim(merged)[1]), ~strata.sims.pop)
merged<-cbind(merged, strata.sims.pop)

allmort.starting.point<-which(names(merged)=="V1")
allmort.ending.point<-which(names(merged)==paste("V", nsim1, sep=""))
totalmort.starting.point<-which(names(merged)=="X1")
totalmort.ending.point<-which(names(merged)==paste("X", nsim1, sep=""))
pop.starting.point<-which(names(merged)=="Y1")
pop.ending.point<-which(names(merged)==paste("Y", nsim1, sep=""))

#RE.PAFs.info<-merged[,c(unlist(strata.combos[j]), "outcome"), with=FALSE]
RE.PAFs<-cbind(merged[,c("outcome", "riskfactor"), with=FALSE],
              merged[,allmort.starting.point:allmort.ending.point]/merged[,totalmort.starting.point:totalmort.ending.point])

setorder(RE.PAFs, riskfactor, outcome)
sum.stats.attr.mort.PAF<-sum.stats.maker(strata.sims.allmort)
RE.PAFs.sum.stats<-sum.stats.maker.RE.PAFs(RE.PAFs)

standardized.mort<-cbind(merged[,c("outcome", "riskfactor"), with=FALSE],
                        merged[,allmort.starting.point:allmort.ending.point]/merged[,pop.starting.point:pop.ending.point])
setorder(standardized.mort, riskfactor, outcome)

standardized.mort.sum.stats<-sum.stats.maker(standardized.mort)
names(standardized.mort.sum.stats)[(length(names(standardized.mort.sum.stats))-4):length(names(standardized.mort.sum.stats))]=
  paste("mort.per.100k.", names(standardized.mort.sum.stats)[(length(names(standardized.mort.sum.stats))-4):length(names(standardized.mort.sum.stats))])

#merged.results<-merge(sum.stats.attr.mort.PAF, RE.PAFs.sum.stats)
merged.results<-reduce(list(sum.stats.attr.mort.PAF, RE.PAFs.sum.stats, standardized.mort.sum.stats), merge)

strata.summary.stats<-merged.results

return(list(strata.summary.stats, strata.sims.allmort, strata.sims.totalmort, RE.PAFs, standardized.mort))

}

```

The next function, “sum.by.strata.join”, is the new and improved version of a function to sum joint effects by subgroups of interest, taking in joint attributable mort sims (allmort), total observed mort sims (total mort), population count sims (pop), and a string vector for your strata of interest (covar). It starts out with the usual: converting inputs to data frames and identifying column names for the sims. Note that allmort is going to be sims for joint attributable mortality, not the file with attributable mort sims for each dietary factor.

```
Sum.by.strata.joint <- function(allmort, totalmort, pop, covar)
{
  allmort.dt<-as.data.table(allmort)
  totalmort.dt<-as.data.table(totalmort)
  pop.dt<-as.data.table(pop)

  cols.to.sum.allmort<-paste("V", 1:n.sims, sep="")
  cols.to.sum.totalmort<-paste("X", 1:n.sims, sep="")
  cols.to.sum.pop<-paste("Y", 1:n.sims, sep="")
}
```

Go ahead and sum joint attributable mortality sims by strata of interest.

```
strata.sims.allmort<-allmort.dt[,lapply(.SD, sum), by=c(covar), .SDcols=cols.to.sum.allmort]
```

PAF.draws has already been read in (it's the joint PIF sims for all strata). We take this and, depending on what's "covar" just take "outcome" and whichever "disease\_type" column is of interest (if at all). Then merge by "outcome" with total mort sims. What we're doing here is adding the "disease\_type" variable of interest so that it exists in the total mort sims file.

```
# need to merge totalmort.dt with disease type var

if("disease_type" %in% covar){

  foo <- PAF.draws %>% select(outcome, disease_type) %>% distinct()
  totalmort.dt1 <- left_join(totalmort.dt, foo, by = "outcome")

} else if ("disease_type1" %in% covar){

  foo <- PAF.draws %>% select(outcome, disease_type1) %>% distinct()
  totalmort.dt1 <- left_join(totalmort.dt, foo, by = "outcome")

} else if ("disease_type2" %in% covar){

  foo <- PAF.draws %>% select(outcome, disease_type2) %>% distinct()
  totalmort.dt1 <- left_join(totalmort.dt, foo, by = "outcome")

} else if ("disease_type3" %in% covar){

  foo <- PAF.draws %>% select(outcome, disease_type3) %>% distinct()
  totalmort.dt1 <- left_join(totalmort.dt, foo, by = "outcome")

} else {

  totalmort.dt1 <- totalmort.dt
  print("no change needed")

}
```

Sum by strata of interest for total mortality sims and population count sims.

```

# this worked last time I edited code
# update: we want this to have disease_type3
strata.sims.totalmort <- totalmort.dt1[,lapply(.SD, sum),
                                           by=c(covar[covar %in% c("age", "female", "race", "outcome",
                                           # "pathway",
                                           "disease_type", "disease_type1", "disease_type2
                                           )]),
                                           .SDcols=cols.to.sum.totalmort]

strata.sims.pop <- pop.dt[,lapply(.SD, sum), by=c(covar[covar %in% c("age", "female", "race")]), .SDcols=c

```

Merge allmort, totalmort pop (all summed by strata of interest) together (different methods on how to do it depending on the number of dimensions for strata.sims.pop and strata.sims.totalmort, hence the if statements).

```

if(length(covar[covar %in% c("age", "female", "race")])>0 & dim(strata.sims.totalmort)[1]>1)
{
  merged <- merge(merge(strata.sims.allmort, strata.sims.totalmort),
                  strata.sims.pop, by=covar[covar %in% c("age", "female", "race")])
}

if(length(covar[covar %in% c("age", "female", "race")])>0 & dim(strata.sims.totalmort)[1]==1)
{
  merge.a <- cbind(strata.sims.allmort, strata.sims.totalmort)
  merged <- merge(merge.a, strata.sims.pop, by=covar[covar %in% c("age", "female", "race")])
}

if(length(covar[covar %in% c("age", "female", "race")])==0 & dim(strata.sims.totalmort)[1]>1)
{
  merged <- cbind(merge(strata.sims.allmort, strata.sims.totalmort), strata.sims.pop)
}

if(length(covar[covar %in% c("age", "female", "race")])==0 & dim(strata.sims.totalmort)[1]==1)
{
  merged <- cbind(strata.sims.allmort, strata.sims.totalmort, strata.sims.pop)
}

```

Identify variables names for attributable mort sims, total mort sims, pop sims, so we can calculate “Reverse\_engineered” PIFs, and standardized attributable mortality. We’re also getting summary stats for attributable mort, RE PIFs, and standardized mort.

```

allmort.starting.point<-which(names(merged)== "V1")
allmort.ending.point<-which(names(merged)==paste("V", nsim1, sep=""))
totalmort.starting.point<-which(names(merged)== "X1")
totalmort.ending.point<-which(names(merged)==paste("X", nsim1, sep=""))
pop.starting.point<-which(names(merged)== "Y1")
pop.ending.point<-which(names(merged)==paste("Y", nsim1, sep=""))

RE.PAFs<-cbind(merged[,c(covar), with=FALSE],
               merged[,allmort.starting.point:allmort.ending.point]/merged[,totalmort.starting.point:totalmort.ending.point])

setorderv(RE.PAFs, cols=c(covar))

sum.stats.attr.mort.PAF<-sum.stats.maker(strata.sims.allmort)
RE.PAFs.sum.stats<-sum.stats.maker.RE.PAFs(RE.PAFs)

standardized.mort<-cbind(merged[,c(covar), with=FALSE],
                        merged[,allmort.starting.point:allmort.ending.point]/merged[,pop.starting.point:pop.ending.point])

setorderv(standardized.mort, cols=c(covar))
standardized.mort.sum.stats<-sum.stats.maker(standardized.mort)
names(standardized.mort.sum.stats)[(length(names(standardized.mort.sum.stats))-4):length(names(standardized.mort.sum.stats))]=
  paste("mort.per.100k.",names(standardized.mort.sum.stats)[(length(names(standardized.mort.sum.stats))-4):length(names(standardized.mort.sum.stats))])

```

Merge all the summary stats output together, and return a list of five files: 1. aforementioned summary stats output for joint effects (this is what you'll be looking at in terms of material you'd actually present in **your paper**), 2. joint attributable mortality sims, 3. total observed mortality sims, 4. "Reverse engineered" joint PIF sims, and 5. standardized joint attributable mortality sims.

```

merged.results<-reduce(list(sum.stats.attr.mort.PAF, RE.PAFs.sum.stats, standardized.mort.sum.stats), merge)

strata.summary.stats<-merged.results

return(list(strata.summary.stats, strata.sims.allmort, strata.sims.totalmort, RE.PAFs, standardized.mort))
}

```